Name of Course	**Bachelor of Computer Application (BCA)**
Semester	**V**
Name of Subject	**Mobile Application Development**
Subject Code	**BCA-503**
Marks	**75 Marks**
Lectures	**50 Lectures**

Unit I

1. Introduction to Android

1.1 Introduction to Android platform and Achitecture
1.2 Basic components of Android, activity life cycle
1.3 Features of ART & Delvik Virtua machine
1.4 Android Application Structure, device screen size compatibility
1.5 Android emulator, working with AndroidManifest.xml

1.1 Introduction to Android platform and Achitecture:

Here's a concise explanation for **"Introduction to Android Platform and Architecture"**:

Introduction to Android Platform:

Android is an open-source mobile operating system developed by **Google**, based on the **Linux kernel**. It is primarily designed for **touchscreen devices** like smartphones and tablets. Android provides a rich application framework that allows developers to build innovative apps and games using Java, Kotlin, or C++. It supports millions of apps available on the **Google Play Store**, making it the most widely used mobile OS globally.

Android Architecture:

Android architecture is divided into **four main layers**:

1. Linux Kernel:

The foundation layer that handles core system services like security, memory management, process management, and hardware drivers.

2. Libraries & Android Runtime:

Libraries provide core functionalities like SQLite, WebKit, and OpenGL.
Android Runtime (ART) runs applications and includes core Java libraries.

3. Application Framework:

Offers high-level services to apps, such as activity management, content providers, and view systems.

4. Applications:

The top layer where user-installed and built-in apps like contacts, messages, and browser reside.

1.2 Basic components of Android, activity life cycle:

Here’s a concise answer on **Basic Components of Android and Activity Lifecycle**:

Basic Components of Android:

Android applications are made up of **four main components**:

1. Activities:

Represent a single screen with a user interface (e.g., a login screen).

2. Services:

Run in the background to perform long-running tasks (e.g., music player).

3. Broadcast Receivers:

Handle communication between Android OS and applications (e.g., battery low alert).

4. Content Providers:

Manage data sharing between applications (e.g., contacts app).

Activity Lifecycle:

An **Activity** goes through different states managed by lifecycle methods:

1. onCreate():

Called when the activity is first created (initial setup).

2. onStart():

Activity becomes visible to the user.

3. onResume():

Activity starts interacting with the user.

4. onPause():

Called when activity is partially hidden.

5. onStop():

Activity is no longer visible.

6. onRestart():

Called after stop, before starting again.

7. onDestroy():

Final call before the activity is destroyed.

1.3 Features of ART & Dalvik Virtual machine:

Here's a concise explanation on the **Features of ART and Dalvik Virtual Machine**:

Dalvik Virtual Machine (DVM):

Designed for Android:

Optimized for mobile devices with limited memory and CPU.

Registers-based:

Uses a register-based architecture (unlike Java's stack-based VM).

.dex Format:
Executes `.dex` (Dalvik Executable) files, which are optimized for low memory usage.

JIT (Just-In-Time) Compilation:
Converts bytecode to machine code during runtime, which can slow performance.

Multiple Instances:
Allows running multiple instances efficiently for multitasking.

Android Runtime (ART):

Replaced Dalvik from Android 5.0 (Lollipop):
Offers better performance and efficiency.

AOT (Ahead-Of-Time) Compilation:
Compiles apps to native code during installation, improving speed and battery life.

Better Garbage Collection:
Reduces pause time and improves memory management.

Improved Debugging:
Offers better debugging and profiling features for developers.

Backward Compatibility:
Supports older apps designed for Dalvik.

1.4 Android Application Structure, device screen size compatibility:

Here's a clear and concise explanation on **Android Application Structure and Device Screen Size Compatibility**:

Android Application Structure:

An Android app project typically contains the following components:

1. Manifest File (`AndroidManifest.xml`):
Declares app components, permissions, and app-level settings.

2. Java/Kotlin Source Code:
Contains the logic of the app using activities, fragments, and services.

3. Resources (`res/` folder):
`layout/`: XML UI layouts
`drawable/`: Images and graphics
`values/`: Strings, styles, dimensions, colors

4. Assets Folder:
Stores raw files like fonts, text files, or HTML.

5. Gradle Files:
Build scripts that manage app dependencies and project configurations.

Device Screen Size Compatibility:

To ensure apps run well on different screen sizes and resolutions, Android provides:

1. **Flexible Layouts:**
   Use **ConstraintLayout**, **LinearLayout**, etc., to adapt to screen changes.

2. **Multiple Layout Resources:**
   Create different layouts using folders like `layout-large`, `layout-sw600dp` for tablets.

3. **Density Buckets:**
   Use drawable folders like `drawable-hdpi`, `drawable-xhdpi` to support various screen densities.

4. **dp and sp Units:**
   Use **dp (density-independent pixels)** for layout sizing and **sp (scale-independent pixels)** for text to ensure consistency across devices.

1.5 Android emulator, working with AndroidManifest.xml:

Here's a crisp and clear answer on **Android Emulator** and **Working with AndroidManifest.xml**:

Android Emulator:

The **Android Emulator** is a virtual device that mimics an Android phone or tablet on your computer.

It allows developers to **test and debug apps** without using a physical device.

Supports different **device configurations**, screen sizes, Android versions, and hardware features.

Part of **Android Studio**, it provides tools like GPS simulation, call/SMS testing, and rotation controls.

Helps in testing **UI responsiveness, performance, and compatibility** on multiple virtual devices.

Working with AndroidManifest.xml:

The **AndroidManifest.xml** is a crucial file in every Android project that provides essential information to the Android system.

Key elements include:

1. **App Components Declaration:**
   Declares all activities, services, broadcast receivers, and content providers.

2. **Permissions:**
   Requests access to features like internet, camera, storage, etc.
   Example: `<uses-permission android:name="android.permission.INTERNET" />`

3. **App Metadata:**
   Defines app name, icon, theme, and launcher activity.

4. **Intent Filters:**
   Specifies how the app can be launched or which actions it can handle.

5. Minimum SDK & Target SDK:

Sets the minimum and target Android versions for app compatibility.